

# HEALTHCONNECT APPOINTMENT EXPERIENCE ANALYSIS

## AnalystLab Africa – Week 4 Assignment

*HealthConnect Appointment Data*

## 1. Dataset Overview

The HealthConnect Appointment dataset contains 5,000 appointment records across 18 columns. The dataset provides information that can be used to understand appointment attendance, missed appointments, reminder activity, booking behaviour, patient characteristics, distance to the clinic, and waiting time.

The dataset contains a combination of numerical, categorical, and date-related variables. These variables provide a suitable foundation for examining appointment patterns and identifying factors that may be associated with patient attendance and the overall appointment experience.

## 2. Data Quality Assessment

A data quality assessment was conducted on the HealthConnect Appointment dataset to determine whether the data is complete, accurate, consistent, and suitable for further analysis. The assessment focused on duplicate records, missing values, data types, and the suitability of the variables for answering the business questions.

The dataset contains 5,000 appointment records across 18 columns. No duplicate records were identified, which suggests that there are no obvious repeated appointment records within the dataset.

However, some missing values were identified. The `reminder_channel` column contains 1,366 missing values, representing the largest proportion of missing data. The `distance_to_clinic_km` column has 90 missing values, while `waiting_time_minutes` has 60 missing values. These missing values should be investigated before analysis because they may affect calculations and the interpretation of results.

The missing values in `reminder_channel` require particular attention. A blank value may indicate that a patient did not receive a reminder rather than that the information was simply not recorded. Therefore, these values should not automatically be replaced with zero. Instead, they should be treated according to the meaning of the field and the available business context.

The numerical variables, including `distance_to_clinic_km` and `waiting_time_minutes`, are appropriate for statistical analysis such as averages, medians, and comparisons across appointment groups. Categorical variables such as gender, appointment status, and reminder channel can also be used to identify patterns in attendance and patient experience.

Overall, the dataset is considered suitable for analysis. The absence of duplicate records is a positive quality indicator, while the main data quality concern is the presence of missing values. These values should be appropriately handled and documented before the final analysis to ensure that the findings are reliable and do not lead to misleading conclusions.

## 3. Business Questions

The analysis will focus on understanding the factors that influence appointment attendance and the overall patient experience at HealthConnect. The following business questions will guide the analysis:

1. What percentage of scheduled appointments are attended, cancelled, or missed (no-shows)?

2. Which factors are most strongly associated with appointment no-shows? This will examine factors such as appointment lead time, reminder channel, patient demographics, and distance to the clinic.
3. Does the number of days between booking and the appointment affect the likelihood of a patient attending the appointment?
4. How does the use of appointment reminders affect patient attendance and no-show rates?
5. Does distance from the clinic influence appointment attendance or no-show behaviour?
6. Which patient groups have the highest appointment attendance and no-show rates? This will compare relevant demographic groups to identify patterns in appointment behaviour.
7. What is the average patient waiting time, and which factors are associated with longer waiting times?
8. Which appointment types or clinic services experience the highest waiting times or no-show rates?
9. What areas of the appointment process could HealthConnect improve to reduce no-shows and improve the patient experience?

These questions will help HealthConnect identify patterns in appointment attendance, understand potential causes of missed appointments, evaluate the effectiveness of reminders, and identify opportunities to improve operational efficiency and patient experience.

## 4. Proposed KPIs and Justification

Based on the business questions and the objectives of the HealthConnect appointment analysis, the following five Key Performance Indicators (KPIs) are proposed:

### 1. Appointment Attendance Rate (%)

**Definition:** The percentage of scheduled appointments that patients attended.

**Justification:** This KPI provides a direct measure of appointment attendance and helps HealthConnect understand how effectively patients are attending their scheduled appointments. Monitoring this measure over time and across patient groups can help identify areas where attendance needs to be improved.

### 2. Appointment No-Show Rate (%)

**Definition:** The percentage of scheduled appointments where patients did not attend and did not cancel.

**Justification:** Missed appointments can reduce clinic efficiency and result in unused appointment slots. Tracking the no-show rate will help HealthConnect identify the scale of the problem and investigate factors that may contribute to missed appointments.

### 3. Reminder Effectiveness (%)

**Definition:** The attendance rate among patients who received appointment reminders compared with those who did not receive reminders.

**Justification:** This KPI will help HealthConnect evaluate whether its reminder process is associated with improved appointment attendance. The results can support decisions about which reminder channels should be prioritised.

#### 4. Average Waiting Time (Minutes)

**Definition:** The average number of minutes patients wait before receiving service.

**Justification:** Waiting time is an important measure of the patient experience and operational efficiency. Monitoring this KPI can help HealthConnect identify services, appointment types, or periods with unusually long waiting times and determine where improvements may be required.

#### 5. Average Booking Lead Time (Days)

**Definition:** The average number of days between the date an appointment is booked and the scheduled appointment date.

**Justification:** This KPI provides insight into how far in advance patients are booking appointments. Comparing booking lead time with attendance and no-show rates can help HealthConnect determine whether longer waiting periods between booking and appointment dates are associated with missed appointments.

Together, these KPIs provide a balanced view of patient attendance, missed appointments, reminder effectiveness, waiting experience, and booking behaviour. They will help HealthConnect measure current performance, identify potential causes of appointment issues, and support data-driven recommendations for improving the overall appointment experience.

### 5. Initial Analysis Approach

The initial analysis will focus on identifying patterns and relationships within the HealthConnect appointment data that can help answer the defined business questions. The analysis will be carried out using data cleaning, descriptive analysis, segmentation, and comparative analysis.

First, the dataset will be reviewed and cleaned by checking data types, missing values, inconsistent entries, and duplicate records. Missing values will be assessed based on the meaning of each variable before deciding whether they should be excluded, categorised, or otherwise handled.

Next, descriptive analysis will be conducted to establish the overall performance of HealthConnect appointments. This will include calculating the appointment attendance rate, no-show rate, average waiting time, average booking lead time, and other relevant summary statistics.

The analysis will then compare appointment outcomes across important factors such as reminder channel, booking lead time, distance to the clinic, patient demographics, and appointment type. This will help identify which factors may be associated with higher attendance, higher no-show rates, or longer waiting times.

For the reminder analysis, appointment attendance and no-show rates will be compared between patients who received reminders and those who did not. Different reminder channels will also be compared where sufficient data is available.

Booking lead time and distance to the clinic will be grouped into meaningful categories to make patterns easier to identify. For example, appointment lead time can be grouped into short, medium, and long booking periods, while distance can be grouped into suitable distance ranges.

Waiting time will also be analysed across appointment types and relevant patient groups to identify areas where patients experience longer delays.

Finally, the findings will be presented using appropriate tables, charts, and KPI measures. The analysis will focus on identifying meaningful trends and relationships rather than assuming that one variable directly causes another. The insights generated will be used to develop practical recommendations for improving appointment attendance, reducing no-shows, and enhancing the overall patient experience.

## 6. Assumptions, Limitations, Risks and Dependencies

### 6.1 Assumptions

- The appointment records are assumed to represent genuine HealthConnect appointments.
- Each row is assumed to represent one appointment record.
- Appointment status is assumed to accurately indicate whether an appointment was attended, cancelled, or missed.
- A missing reminder channel is assumed to require further investigation before determining whether it represents no reminder being sent or missing information.
- The available variables are assumed to be sufficient to identify the major factors influencing appointment attendance and patient experience.

### 6.2 Limitations

- The dataset contains missing values, particularly in the `reminder_channel` column, which may limit some aspects of the analysis.
- The dataset provides appointment-level information but may not capture all factors that influence patient attendance, such as personal circumstances, health conditions, or unexpected events.
- The analysis can identify relationships and patterns within the data but cannot establish that one factor directly causes another.
- The dataset represents a specific period of appointments, so the findings may not necessarily represent long-term behaviour.

### 6.3 Risks

- Incorrect interpretation or handling of missing values could result in misleading findings.
- Outliers in variables such as waiting time or distance to the clinic could influence averages and affect the interpretation of results.
- Inconsistent or incorrectly recorded appointment information could reduce the reliability of the analysis.
- Over-reliance on a single KPI or variable may provide an incomplete view of the patient experience.

### 6.4 Dependencies

- Accurate and complete appointment records are required for reliable analysis.

- Clear definitions of appointment statuses, reminder channels, and other key fields are necessary for consistent interpretation.
- Appropriate handling of missing values and outliers is dependent on understanding the business meaning of the affected variables.
- Access to relevant business context from HealthConnect would improve the interpretation of the findings and recommendations.

## 7. Project Summary

The HealthConnect Appointment Experience analysis aims to understand appointment attendance, identify factors associated with missed appointments, evaluate the effectiveness of appointment reminders, and assess the overall patient experience.

The analysis will use the HealthConnect appointment dataset to examine key measures such as appointment attendance rate, no-show rate, reminder effectiveness, average waiting time, and booking lead time. Different patient groups, reminder channels, booking periods, distances, and appointment types will be compared to identify meaningful patterns.

The analysis will also assess the quality of the available data and appropriately address missing values and other potential data-quality issues. The findings will be presented through clear KPIs, visualisations, and insights that can support evidence-based decision-making.

Ultimately, the project is expected to help HealthConnect identify opportunities to reduce appointment no-shows, improve appointment management, evaluate reminder strategies, reduce waiting times, and enhance the overall patient experience.